

Please write clearly in block capitals.

Centre number

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Candidate number

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Surname _____

Forename(s) _____

Candidate signature _____

I declare this is my own work.

INTERNATIONAL AS CHEMISTRY (9620)

Unit 1: Inorganic 1 and Physical 1

Friday 5 January 2024

07:00 GMT

Time allowed: 1 hour 30 minutes

Materials

For this paper you must have:

- the Periodic Table/Data Sheet, provided as an insert
- a ruler with millimetre measurements
- a scientific calculator, which you are expected to use where appropriate.

Instructions

- Use black ink or black ball-point pen.
- Fill in the boxes at the top of this page.
- Answer **all** questions.
- You must answer the questions in the spaces provided. Do **not** write outside the box around each page or on blank pages.
- If you need extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
- All working must be shown.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 70.

For Examiner's Use	
Question	Mark
1	
2	
3	
4	
5	
6	
7	
TOTAL	



Answer **all** questions in the spaces provided.

0 1

Gallium (Ga) is a metal.

0 1 . 1

Describe the bonding in solid gallium.

[2 marks]

Gallium forms GaCl_3 molecules.

0 1 . 2

Write an equation for the formation of GaCl_3 from its elements.

[1 mark]

0 1 . 3

Draw the shape of a molecule of GaCl_3

Include any lone pairs of electrons, if any, that influence the shape.

Name the shape of the GaCl_3 molecule.

[2 marks]

Shape

Name of shape _____



0 1 . 4 A molecule of GaCl_3 reacts with a chloride ion to form a GaCl_4^- ion.

During this reaction a bond is formed.

Name the type of bond formed.

Describe how this bond is formed.

[2 marks]

Type of bond _____

How bond is formed _____

0 1 . 5 What is the difference, if any, between the Cl–Ga–Cl bond angle in GaCl_3 and in GaCl_4^- ?

Tick (✓) **one** box.

[1 mark]

The bond angle is bigger in GaCl_3

☐

The bond angle is the same

☐

The bond angle is smaller in GaCl_3

☐

8

Turn over for the next question

Turn over ►



0 2

This question is about reactions involving halide ions.

0 2 . 1

Give a reagent that could be used to distinguish between separate samples of aqueous potassium fluoride and aqueous potassium chloride.

Describe what you would observe for each solution.

[3 marks]

Reagent _____

Observation with aqueous potassium fluoride _____

Observation with aqueous potassium chloride _____

Aqueous bromine reacts with aqueous sulfur dioxide to form hydrobromic acid (HBr).

0 2 . 2

Write a half-equation for the reaction of bromine to form bromide ions.

[1 mark]

0 2 . 3

Write a half-equation for the reaction of an aqueous solution of sulfur dioxide to form hydrogen ions and sulfate(VI) ions.

[1 mark]

0 2 . 4

Use your answers to Question **02.2** and Question **02.3** to deduce an overall equation for the reaction of bromine and aqueous sulfur dioxide to form hydrobromic acid.

[1 mark]

0 2 . 5

Identify the reducing agent in this reaction.

[1 mark]



0	2	.	6
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Chlorine gas is bubbled into an aqueous solution of potassium iodide.

Give **one** observation.

Write an equation for this reaction.

[2 marks]

Observation _____

Equation _____

9

Turn over for the next question

Turn over ►



0 3

Time of flight (TOF) mass spectrometry is used to determine accurate relative atomic masses.

0 3 . 1

Define relative atomic mass.

[2 marks]

A sample of germanium (Ge) is ionised by electron impact.

0 3 . 2

Write an equation, including state symbols, for this ionisation.

[1 mark]



0 3 . 3

In a TOF mass spectrometer, some germanium ions are accelerated to a kinetic energy (KE) of $9.616 \times 10^{-13} \text{ J}$

One germanium ion takes $3.739 \times 10^{-7} \text{ s}$ to travel along a 1.500 m flight tube.

$$KE = \frac{mv^2}{2}$$

m = mass of the ion / kg

v = velocity of ion / m s^{-1}

Calculate the mass, in g, of one mole of these germanium ions.

Use your answer to deduce the mass number of this germanium ion.

The Avogadro constant, $L = 6.022 \times 10^{23} \text{ mol}^{-1}$

[5 marks]

Mass of one mole of germanium ions _____ g

Mass number _____

0 3 . 4

Explain why the atomic radius of ^{76}Ge is the same as the atomic radius of ^{74}Ge

[1 mark]

9

Turn over for the next question

Turn over ►



0 4

This question is about compounds containing carbon.

0 4 . 1

Deduce the formula of silicon carbide.

[1 mark]

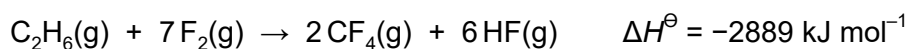
0 4 . 2

The C–F bond is polar.

Explain why CF_4 is a non-polar molecule.

[2 marks]

0 4 . 3

Ethane (C_2H_6) reacts with fluorine to form CF_4 **Table 1** gives some values of standard enthalpies of formation, $\Delta_f H^\ominus$ **Table 1**

Substance	$\text{C}_2\text{H}_6(\text{g})$	$\text{F}_2(\text{g})$	$\text{HF}(\text{g})$
$\Delta_f H^\ominus / \text{kJ mol}^{-1}$	–85	0	–269

Use the standard enthalpy change of the reaction and the data in **Table 1** to calculate the standard enthalpy of formation, in kJ mol^{-1} , of $\text{CF}_4(\text{g})$

[3 marks]

Standard enthalpy of formation of $\text{CF}_4(\text{g})$ _____ kJ mol^{-1} 

0	4
---	---

.

4

 Methanol (CH_3OH) dissolves in water.

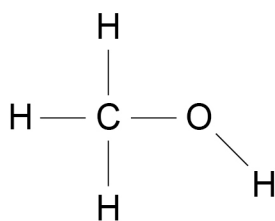
Name the strongest type of interaction between a molecule of methanol and a molecule of water.

Complete the diagram to show how one molecule of methanol interacts with one molecule of water.

Include all lone pairs and partial charges in your diagram.

[4 marks]

Name _____



0	5
---	---

This question is about a titration experiment.

0	5	.	1
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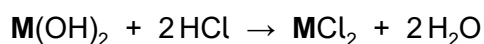
A student does a titration to identify a metal, **M**.

The student dissolves 2.43 g of the metal hydroxide, **M**(OH)₂, in distilled water to make 250 cm³ of an aqueous solution.

A 25.0 cm³ sample of this **M**(OH)₂ solution is titrated with 0.104 mol dm⁻³ hydrochloric acid.

The mean titre is 27.30 cm³

The equation for the reaction is



Calculate the relative atomic mass of **M**.

Give your answer to 1 decimal place.

Identify **M**.

[6 marks]

Relative atomic mass of **M** _____

Identity of **M** _____



The student makes the solution of the metal hydroxide in Question **05.1** as follows.

- Step 1** Weighs out the solid metal hydroxide and records the mass to 2 decimal places
- Step 2** Transfers the solid to a beaker
- Step 3** Dissolves the solid in distilled water
- Step 4** Transfers the solution, with washings from the beaker, to a suitable piece of equipment
- Step 5** Makes up the volume of solution with distilled water to exactly 250 cm³

0 5 . 2

Describe what the student must do to make sure they know the exact mass of the solid metal hydroxide in **Step 1**.

[1 mark]

0 5 . 3

Name a suitable piece of equipment the student would use in **Step 4**.

[1 mark]

0 5 . 4

Describe how the student would make the solution up to exactly 250 cm³ in **Step 5**.

[1 mark]

Question 5 continues on the next page

Turn over ►



0 5 . 5 Table 2 shows the results of the experiment.

Table 2

	Rough	1	2	3
Final reading / cm³	29.20	29.55	29.35	30.95
Initial reading / cm³	1.50	2.20	1.85	3.70
Titre / cm³	27.70	27.35	27.50	27.25

Explain why the mean titre used in the calculation in Question **05.1** is 27.30 cm³

[1 mark]

0 5 . 6 Distilled water is used to wash down the sides of the conical flask near the end of the titration.

Give **one** reason why this washing is done.

Give **one** reason why distilled water is used.

[2 marks]

Reason for washing down the sides of the flask _____

Reason why distilled water is used _____



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0	6
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This question is about the elements in Group 2, from Mg to Ba, and their compounds.

0	6	.	1
---	---	---	---

State why the atomic radius of barium is larger than the atomic radius of calcium.

[1 mark]

0	6	.	2
---	---	---	---

Strontium reacts with hydrochloric acid.

When 0.123 g of strontium is added to hydrochloric acid, a gas is produced.

Write an equation for the reaction of strontium with hydrochloric acid.

Calculate the volume, in cm^3 , of the gas produced at 101 kPa and 25 °C
Give your answer to 3 significant figures.

The gas constant, $R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$

[6 marks]

Equation _____

Volume _____ cm^3



Two Group 2 metals, **Y** and **Z**, form chlorides **YCl₂** and **ZCl₂**
A student does some test tube reactions with solutions of **YCl₂** and **ZCl₂**
The observations are shown in **Table 3**.

Table 3

	Addition of potassium hydroxide solution	Addition of potassium sulfate solution
YCl₂	Colourless solution	White precipitate
ZCl₂	White precipitate	Colourless solution

0 6 . 3 Identify metal **Y** and metal **Z**.

[2 marks]

Metal **Y** _____ Metal **Z** _____

0 6 . 4 Write equations to show the formation of each of the white precipitates.

If you were unable to identify the metals **Y** and **Z**, you should use **YCl₂** and **ZCl₂** in your equations.

[2 marks]

Equation 1

Equation 2

0 6 . 5 A student dissolves 2.53 g of calcium chloride in water at 20.0 °C in a plastic cup. 50.0 cm³ of solution are formed and the temperature of the solution rises to 25.4 °C

Calculate a value for the enthalpy change, in kJ mol⁻¹, when calcium chloride dissolves in water.

Specific heat capacity of the solution is 4.18 J g⁻¹ K⁻¹
Assume that the density of the solution is 1.00 g cm⁻³

[3 marks]

Enthalpy change _____ kJ mol⁻¹

14**Turn over ►**

0 7

This question is about physical properties of elements in Period 3.

0 7 . 1

Identify the element in Period 3 with the lowest melting point.

[1 mark]

0 7 . 2

State the meaning of electronegativity.

[1 mark]

0 7 . 3

Suggest why the electronegativity of the elements in Period 3 increases from sodium to chlorine.

[2 marks]

0 7 . 4

Table 4 shows successive ionisation energies of an element in Period 3.

Table 4

Ionisation number	1	2	3	4	5	6	7	8
Ionisation energy / kJ mol ⁻¹	1060	1900	2920	4960	6280	21 200	25 900	30 500

Identify this element.

[1 mark]



0 7 . 5

Write an equation, including state symbols, to show the process that occurs when the **second** ionisation energy of sodium is measured.

[1 mark]

0 7 . 6

Explain why sodium has a higher **second** ionisation energy than magnesium.

[2 marks]

8

END OF QUESTIONS



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